ShopSphere: Project Presentation & Meeting Script

Full-Stack E-Commerce Platform with End-to-End DevOps Architecture
Team Members: Shreya Khandelwal (Lead), Khushboo Lakhani, Mahek
Meeting Duration: ~8–10 Minutes | **Date/Time:** Today at 2:00 PM

1. Quick Project Summary (The Elevator Pitch)

> **ShopSphere** is a modern, enterprise-ready full-stack E-Commerce platform engineered to simulate real-world software delivery. It features a responsive **React 18** frontend, a scalable **Node.js/Express** backend, relational persistence via **PostgreSQL/SQLite**, orchestrated using **Docker Compose**, continuously integrated through **Jenkins CI/CD** & **GitHub Actions**, planned using **Jira Agile workflows**, and monitored with **Prometheus & Grafana**.

2. Team Work Breakdown & Responsibilities

Team Member	Role	Core Deliverables & Technical Ownership	Git Branch & Jira Keys
Shreya Khandelwal	**Team Lead & Platform / DevOps Architect**	• Overall full-stack architecture & data modeling (Products, Orders) • Container orchestration with Docker & multi-service Docker Compose • Declarative Jenkins CI/CD pipeline automation & GitHub Actions CI • Prometheus metrics instrumentation (`/metrics`) & Grafana telemetry • Repository governance, PR code reviews, branch merging, and Jira Sprint management	`main` , `develop` `SHOP-101` , `SHOP-102` , `SHOP-103` , `SHOP-203` , `SHOP-204`
Khushboo Lakhani	**Frontend & User Experience Engineer**	• Product Reviews & Customer Star Ratings component (`SHOP-201`) • Review submission modal, validation, and real-time average score recalculation • Responsive styling and product feedback badge display • Feature branching, Git smart commits, and peer Pull Request lifecycle	`feature/khushboo-product-reviews` `SHOP-201`
Mahek	**Backend Logic & Promotions Specialist**	• Coupon & Discount Promotion Engine (`SHOP-202`) • Real-time promo code validation (`SAVE20` for 20% off, `DEVOPS10` for $10 off) • Subtotal recalculation in Cart Drawer and final checkout billing • Interactive coupon application badge and discount status feedback	`feature/mahek-discounts-coupons` `SHOP-202`

3. Step-by-Step Speaking Script (8–10 Minutes)

Part 1: Welcome & Executive Overview (1.5 Mins) — Speaker: Shreya

> *"Good afternoon everyone. Today, our team—myself as Lead, along with Khushboo and Mahek—is presenting **ShopSphere**, our full-stack E-Commerce web platform built with enterprise-grade DevOps automation.*
>
> *Our goal with this project was not just to build a functional online store, but to implement the entire modern Software Development Life Cycle (SDLC). That includes Agile sprint planning in Jira, Git branch-per-feature workflows, containerized microservices using Docker Compose, automated CI/CD testing and builds with Jenkins, and live telemetry with Prometheus and Grafana.*
>
> *I'll now hand over to Khushboo to present the customer reviews and rating architecture."*

Part 2: Product Reviews & Customer Experience (2 Mins) — Speaker: Khushboo

> *"Thank you, Shreya. In an E-Commerce platform, customer trust and social proof are critical for conversion. I was responsible for designing and implementing the **Customer Reviews and Star Ratings component** under Jira ticket **SHOP-201**.*
>
> *Here is what I built:*
>
> 1. *An interactive **Star Rating Selector** allowing customers to score products from 1 to 5 stars.*
> 2. *A **Review Submission Modal** that captures customer feedback, validates input, and updates the product's overall rating average and review count.*
> 3. *Visual badges on each product card displaying the verified customer rating score.*
>
> *On the DevOps side, I followed our team Git workflow by creating my own feature branch `feature/khushboo-product-reviews`, committed my work using smart commit syntax linked to SHOP-201, and submitted a Pull Request which was peer-reviewed and merged into our `main` production release branch.*
>
> *Next, Mahek will walk through our promotions and discount engine."*

Part 3: Promotions, Discount Engine & Checkout (2 Mins) — Speaker: Mahek

> *"Thanks, Khushboo. I developed the **Coupon Code and Discount Promotion System**, tracked under Jira user story **SHOP-202**.*
>
> *Key technical highlights of my contribution include:*
>
> 1. *A dynamic **Discount Calculation Engine** integrated into the Shopping Cart Drawer.*
> 2. *Support for multiple promo types: percentage discounts (such as `SAVE20` for 20% off) and flat credits (like `DEVOPS10`).*
> 3. *Instant cart subtotal recalculation, sales tax estimation, and visual discount confirmation badges.*
> 4. *Persistence of the discount in the final order transaction sent to our database upon checkout.*
>
> *Like Khushboo, I developed this in my isolated feature branch `feature/mahek-discounts-coupons`, pushed my commits to GitHub, and merged via Pull Request into the main branch after passing all test suites.*
>
> *Now, Shreya will explain the underlying DevOps, CI/CD pipeline, and telemetry infrastructure."*

Part 4: Cloud Infrastructure, CI/CD & Telemetry (2.5 Mins) — Speaker: Shreya

> *"Thank you, Mahek. To support our store with 99.9% uptime and rapid release velocity, I architected the underlying DevOps platform across four key pillars:*
>
> 1. ***Containerization & Multi-Service Topology:*** *We built multi-stage Dockerfiles for both frontend (using Alpine Nginx) and backend (Node.js). With a single command— `docker compose up --build -d` —all 5 containers spin up on an isolated bridge network with PostgreSQL persistent data volumes.*
> 2. ***Declarative Jenkins CI/CD Pipeline:*** *Our Jenkinsfile pipeline automates 5 consecutive stages: • **Git Checkout** from our GitHub repository (`shreya18633/shopsphere-ecommerce`). • **Code Quality & Linting** to ensure clean standards. • **Automated Unit Testing** via Jest, testing API routes, health endpoints, and database models. • **Docker Build** packaging production-ready images. • **Automated Deployment & Health Verification** on port 5000 and 3000.*
> 3. ***Observability with Prometheus & Grafana:*** *We integrated Prometheus metrics middleware into our backend. It exposes latency histograms, HTTP request counters, total checkouts, and system memory consumption at `/metrics` . This gives us real-time telemetry into cluster performance.*
> 4. ***Agile Planning & Smart Commits:*** *In Jira, we organized 4 core user stories across Sprint 1 and Sprint 2, maintaining complete traceability from requirement ticket to git commit, Jenkins build, and live monitoring gauge."*

Part 5: Live Walkthrough & Wrap-Up (1 Min) — Speaker: Shreya / Team

> *"To conclude our presentation, we would like to showcase the live application:*
>
> *• **Browsing & Categories:** [Demo: Filter by Electronics / Accessories] • **Customer Reviews (Khushboo):** [Demo: Open reviews on Sony headphones] • **Discount & Checkout (Mahek):** [Demo: Add item to cart, apply `SAVE20` , and place order] • **DevOps Console (Shreya):** [Demo: Click 'DevOps Console' in navbar to show Jenkins pipeline simulator, Docker architecture topology, Jira agile sprint matrix, and live Prometheus scrape metrics]*
>
> *Thank you! We are now open for any questions or feedback."*

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4. Anticipated Meeting Questions & Quick Answers

Question	Answer / Quick Response
"How did your team collaborate on Git?"	*"We used feature branching (`feature/khushboo-product-reviews` , `feature/mahek-discounts-coupons`). Neither member committed directly to `main` . All code was merged via GitHub Pull Requests after automated tests passed."*
"What happens if PostgreSQL goes down?"	*"Our database configuration in `db.js` includes an automatic zero-config SQLite fallback, preventing server crashes during local demonstrations or test runs."*
"How does Jenkins ensure code quality?"	*"The pipeline executes Jest tests during Stage 2. If any test fails, the build halts immediately, preventing faulty containers from deploying to production."*
"What business metrics are captured in Prometheus?"	*" `ecommerce_orders_total` (total checkouts), `ecommerce_revenue_total` (gross revenue), and `http_request_duration_seconds` (latency histogram)."*